Jason Cosgrove

PhD · Principal Scientist



Hybrid scientist × engineer. I build agentic AI workflows for biomedical discovery, then validate them in the lab. My goal is safe AI systems for diverse biomedical applications, built by combining safety-critical systems engineering, biology and computational biology.

TECHNICAL SKILLSET

Agentic AI & LLM-driven pipelines

Python (incl. PyTorch), R, Java

Test-driven development, verification & validation

Literature, multi-omics & population-health data

Target, biomarker & asset evaluation

Oncology, immunology & haematology

Wet-lab validation: imaging, cytometry, in vivo / ex vivo models

CONTACT

Paris, France

Full right to work in the UK

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LinkedIn profile

Google Scholar profile

Career Highlights

Evidence-driven target & asset assessment

At Servier I run due-diligence evaluations of external biotech assets with the Business Development team, building target opportunity packages from omics and other data to inform in-licensing decisions.

Agentic AI, validated in the lab

I build proof-of-concept agentic discovery workflows, take their predictions to lab validation, and work with engineering teams to take them to production. Trained in test-driven engineering: in my PhD I developed an acceptance-test-driven method for building multi-agent systems.

Partnerships & consulting

I co-founded C3 Bioscience, which helps start-ups build their computational biology capability. I also manage academic–private partnership projects with Institut Curie and Imperial College London that develop new experimental and computational tools for drug discovery.

Teaching others to use AI

I lead AI training for the British Society for Immunology. I have been a guest lecturer at Oxford, Imperial, Sorbonne, ENS and Shanghai Jiao Tong University.

Experience

2024 – now

Principal Scientist, Computational Medicine *Servier Pharmaceuticals, France*

- Built **end-to-end early-discovery workflows** in which **AI agents assemble the analysis pipelines**, with prioritised targets taken through to **lab validation**.
- Develop **proof-of-concept agentic pipelines** used by internal R&D teams, and work with Servier's AI and engineering teams to industrialise them at production scale.
- **Due-diligence evaluations of external assets** with Business Development: target opportunity packages summarising key omics and other data trends to inform **in-licensing decisions**.
- Computational biology lead for precision-oncology and immuno-oncology projects. Contributed to strategic roadmaps for therapeutic projects and to the group's **Data & AI strategy**.

2025 – now

Co-founder & Principal Consultant *C3 Bioscience*

- Help start-ups build out their computational biology capabilities, working inside their teams as an external technical partner.
- Prepared **pre-seed data packages** for two biotech start-ups, using mechanistic analyses of public datasets to de-risk their programmes for investors.
- Delivered AI and computational biology training at Institut Pasteur.

Now

AI Training Lead *British Society for Immunology*

Lead the Society's AI training initiatives, helping immunologists use AI in their research.

2018 – 2024

Postdoctoral Fellow *Institut Curie, Paris*

- Mined the **NHANES population-health database** with interpretable machine learning (XGBoost) to find non-genetic factors that predict myeloid blood cell counts.
- Identified novel therapeutic targets and biomarkers for immune cell development using multi-omics (single-cell, lineage barcoding, metabolic profiling), with in vivo / ex vivo validation.
- Secured **~€4M in funding** (ERC, Wellcome Trust, Servier Innovation) for an independent research programme.

WHAT I BRING

Pharma R&D experience turning omics and other evidence into go/no-go decisions on targets and external assets, an engineer's test-first discipline, and partner-facing work with biotechs, start-ups and academic centres. Over €4M in funding and papers in *Nature Cell Biology*, *Nature Metabolism* & *Nature Communications*.

SOFTWARE

Agentic AI pipelines for target & biomarker discovery (Servier)

CXCL13Sim: multi-agent immune simulation, built test-first (Java)

SPARTAN: R package for statistical analysis of simulations (contributor)

CellDestiny: R Shiny app for single-cell lineage-tracing data (co-developer)

TEACHING & MENTORING

AI training lead, British Society for Immunology

Guest lecturer: Oxford, Imperial College London, Sorbonne, ENS, Shanghai Jiao Tong

Organised computational biology workshops at ENS & Institut Pasteur

Supervised PhD students and researchers from mathematics, biology, computer science & statistics

REFERENCES

Available upon request

Experience (continued)

2013 – 2017 **PhD Researcher** *University of York, UK*

- Designed, built and validated **CXCL13Sim**, a **multi-agent simulation** of immune cells in lymphoid tissue (Java / MASON).
- Developed a **test-driven approach to building multi-agent systems**: acceptance tests derived from requirements, backed by unit, module and system-level tests in an automated JUnit regression suite with coverage tracking.
- Applied **safety-critical systems engineering** methods (CoSMoS, UML specification, safety-case-style argumentation) to show the simulator was fit for purpose. Quantified uncertainty with sensitivity analysis, ML emulators and multi-objective optimisation.
- **Validated model predictions in the lab** with super-resolution imaging, published in *Nature Communications* (first author).

Education

2013 – 2017 **PhD** *University of York, UK*
Combining engineering, machine learning & imaging approaches to study immune cell niches

2012 – 2013 **MSc, Bioinformatics & Systems Biology** *University College Cork, Ireland*

2008 – 2012 **BSc, Biomedical, Health & Life Sciences** *University College Dublin, Ireland*

Selected Publications

15 peer-reviewed publications. Full list on Google Scholar.

1. **Cosgrove J**, Donada A, Lyne A, et al. (2026) The pentose phosphate pathway regulates myelo-lymphoid lineage specification. *Nature Communications* (accepted).
   Metabolic state-fate mapping (in situ barcoding + metabolomics) shows stem cell metabolism can be targeted to steer immune cell production; validated in vivo.
2. **Cosgrove J**, et al. (2024) A call for accessible tools to unlock single-cell immunometabolism research. *Nature Metabolism*.
   First-author perspective with leading international immunometabolism labs on the tools the field needs.
3. Gerber-Ferder Y, **Cosgrove J**, et al. (2023) Breast cancer remotely imposes a myeloid bias on haematopoietic stem cells by reprogramming the bone marrow niche. *Nature Cell Biology*.
   Shows that tumours reprogramme the bone marrow niche from a distance, pushing stem cells towards pro-tumour myeloid output.
4. **Cosgrove J**, Novkovic M, et al. (2020) B cell zone reticular cell microenvironments shape CXCL13 gradient formation. *Nature Communications*.
   PhD paper: agent-based simulation and optimisation of immune cell dynamics, with predictions validated by imaging and knockout mice.
5. Alden K, **Cosgrove J**, Coles M, Timmis J. (2020) Using emulation to engineer and understand simulations of biological systems. *IEEE/ACM Trans. Comput. Biol. Bioinform.*
   Machine-learning emulator workflow that makes evaluating multi-agent simulations far more efficient (results in seconds), enabling multi-objective optimisation and Bayesian uncertainty analysis; released in the spartan R package.
6. **Cosgrove J**, Butler J, Alden K, et al. (2015) Agent-based modeling in systems pharmacology. *CPT: Pharmacometrics & Systems Pharmacology*.
   Tutorial introducing agent-based modelling to pharmacologists, with preclinical drug-discovery case studies.